

Myeloma Emergencies & Supportive Care (Bone, Ca, Cord, Hyperviscosity)



1. Bone disease — bisphosphonate/denosumab

WHAT YOU SEE

A 'BONE DISEASE BAY' showing osteoclast resorption with denosumab and zoledronate options.

WHAT IT ENCODES

All myeloma patients with bone disease get a bone-modifying agent: zoledronic acid (IV bisphosphonate) or denosumab (RANKL antibody) to reduce skeletal events. Denosumab is preferred in renal impairment (not renally cleared). Both risk osteonecrosis of the jaw — dental evaluation first.

BOARD TRAP

Use DENOSUMAB instead of zoledronate when CrCl is low/renal failure (bisphosphonates are nephrotoxic and renally dosed) — and screen for dental disease to avoid ONJ in both.

2. Acute hypercalcemia — IV saline first

WHAT YOU SEE

An 'ACUTE HYPERCALCEMIA BAY' with hydration/calciuresis, zoledronate, calcitonin, and steroids.

WHAT IT ENCODES

Treat acute hypercalcemia with aggressive IV normal saline to restore volume and promote calciuresis FIRST, then calcitonin for rapid (but transient) lowering and a bisphosphonate (zoledronate) for durable control. Steroids help in myeloma. Dialysis for severe refractory cases.

BOARD TRAP

First step in hypercalcemia is IV NORMAL SALINE volume repletion — NOT a loop diuretic; furosemide is only after euvolemia, and bisphosphonate effect takes 2-4 days.

3. Zoledronate — renal dose / dental check

WHAT YOU SEE

A 'ZOLEDRONATE' label noting fastest bisphosphonate but renal caution and dental clearance.

WHAT IT ENCODES

Zoledronic acid is the fastest, most potent IV bisphosphonate for hypercalcemia and bone protection but is renally cleared (dose-reduce/avoid in renal impairment) and can cause acute kidney injury if infused too fast. Obtain dental clearance to reduce osteonecrosis-of-jaw risk.

BOARD TRAP

Zoledronate requires renal-function-based dosing and a slower infusion; in significant renal impairment switch to denosumab — and always get a pre-treatment dental exam (ONJ).

4. Calcitonin — rapid but tachyphylaxis

WHAT YOU SEE

A 'CALCITONIN — fastest, rapid' note with a tachyphylaxis caution.

WHAT IT ENCODES

Calcitonin lowers calcium within hours (fastest onset) by inhibiting osteoclasts and increasing renal calcium excretion, bridging until the bisphosphonate works. However, tachyphylaxis develops within ~48 hours, so it is only a short-term bridge, not definitive therapy.

BOARD TRAP

Calcitonin acts fastest but only briefly (tachyphylaxis in ~48h) — it bridges to the bisphosphonate's durable effect; don't rely on it for sustained calcium control.

5. Cord compression — ALERT 1

WHAT YOU SEE

An 'EMERGENCY CORRIDOR — ALERT 1: CORD COMPRESSION' with MRI, high-dose dexamethasone, radiation/surgery.

WHAT IT ENCODES

Spinal cord compression (from vertebral lesion/fracture or plasmacytoma) is a neuro-oncologic emergency: back pain with weakness, sensory level, or sphincter dysfunction. Give high-dose dexamethasone immediately and obtain urgent whole-spine MRI, then emergent radiation or surgical decompression.

BOARD TRAP

Give dexamethasone and order emergent MRI the moment cord compression is suspected — don't wait for imaging to start steroids; delay risks permanent paraplegia.

6. Hyperviscosity — ALERT 2 / plasmapheresis

WHAT YOU SEE

An 'ALERT 2: HYPERVISCOSITY' panel with sausage-link retinal veins and a plasmapheresis machine.

WHAT IT ENCODES

Hyperviscosity syndrome (blurred vision, headache, mucosal bleeding, confusion; 'sausage-link' dilated retinal veins on funduscopy) is treated emergently with PLASMAPHERESIS to rapidly remove paraprotein, followed by therapy directed at the clone. Most common with IgM (Waldenstrom) and high IgA.

BOARD TRAP

Emergency treatment of hyperviscosity is PLASMAPHERESIS, not transfusion — and avoid RBC transfusion before pheresis because it raises viscosity and can precipitate decompensation.

7. DVT prophylaxis bay (IMiD VTE)

WHAT YOU SEE

A 'DVT PROPHYLAXIS BAY' with aspirin (low risk) vs warfarin/LMWH/DOAC (high risk) shields.

WHAT IT ENCODES

Because IMiDs (thalidomide/lenalidomide/pomalidomide) plus dexamethasone are highly thrombogenic, give aspirin for low VTE risk and therapeutic LMWH or a DOAC for high risk (prior VTE, immobility, multiple risk factors). Continue throughout IMiD therapy.

BOARD TRAP

VTE prophylaxis is mandatory with IMiD regimens — aspirin (low risk) vs anticoagulation (high risk); omitting prophylaxis is a classic management error.

8. Anemia — EPO / iron / B12

WHAT YOU SEE

An 'ANEMIA BAY' with 'establish cause first', marrow replacement, EPO and nutritional deficiencies.

WHAT IT ENCODES

Myeloma anemia is multifactorial: marrow infiltration, relative EPO deficiency, renal disease, and treatment effect. Establish/treat correctable causes (iron, B12, folate) and use erythropoiesis-stimulating agents for symptomatic anemia with low EPO, alongside disease control. Transfuse as needed (cautiously if hyperviscosity).

BOARD TRAP

Work up reversible causes (iron/B12/folate, renal) before reflexively transfusing — and treat the underlying myeloma; ESAs are adjuncts, not a substitute for disease control.

9. Acyclovir / infection prophylaxis

WHAT YOU SEE

An 'ACYCLOVIR' / dose board with HZ prophylaxis and herpes-zoster note.

WHAT IT ENCODES

Proteasome inhibitors, anti-CD38 antibodies, bispecifics/CAR-T, and transplant raise infection risk: give acyclovir/valacyclovir against VZV reactivation, PJP prophylaxis with high-dose steroids, consider IVIG for recurrent infections with hypogammaglobulinemia, and ensure vaccinations (encapsulated organisms).

BOARD TRAP

Bortezomib/anti-CD38/transplant patients need VZV (acyclovir) prophylaxis; recurrent infections with hypogammaglobulinemia warrant IVIG — supportive antimicrobial prophylaxis is essential, not optional.

10. Response assessment — MRD / immunofixation

WHAT YOU SEE

A 'RESPONSE ASSESSMENT CORNER' with 'IFE negative, BM biopsy normal, M-protein lab', MRD flow noting longer PFS.

WHAT IT ENCODES

Response is graded by reduction in M-protein/FLC: partial response, very good partial response, and complete response (negative serum/urine immunofixation + <5% marrow plasma cells). Measurable residual disease (MRD) negativity by next-gen flow/sequencing predicts longer PFS/OS and is an increasingly important endpoint.

BOARD TRAP

Complete response requires NEGATIVE immunofixation (not just a low SPEP M-spike) plus <5% marrow plasma cells; MRD-negativity is deeper still and predicts the best outcomes.

11. No IV contrast (renal/casts)

WHAT YOU SEE

A 'NO CONTRAST DYE' caution near the hydration/renal section.

WHAT IT ENCODES

Avoid iodinated IV contrast in active myeloma with renal impairment or high light-chain load — it can precipitate cast nephropathy/contrast nephropathy. Maintain hydration, avoid NSAIDs and nephrotoxins, and treat hypercalcemia to protect kidneys.

BOARD TRAP

Withhold IV contrast (and NSAIDs) in myeloma with renal risk — contrast can trigger AKI/cast nephropathy; this is why whole-body LOW-DOSE CT (non-contrast) is preferred for bone imaging.

12. Pain relief / vertebroplasty

WHAT YOU SEE

A 'PAIN RELIEF' icon with collapsed vertebra and cement (vertebroplasty/kyphoplasty).

WHAT IT ENCODES

Painful vertebral compression fractures may be treated with vertebroplasty/kyphoplasty (cement augmentation) plus radiation for focal bone pain or impending fracture/plasmacytoma. Multimodal analgesia avoids NSAIDs (renal) where possible; low-dose radiotherapy controls localized lesions and cord-threatening disease.

BOARD TRAP

Localized bone pain/plasmacytoma responds to low-dose radiation and, for vertebral fractures, kyphoplasty — but avoid NSAIDs for analgesia given the renal risk in myeloma.

13. Denosumab — preferred in renal failure

WHAT YOU SEE

A 'DENOSUMAB 120mg SC monthly' note in the bone-disease bay, flagged for renal use.

WHAT IT ENCODES

Denosumab (anti-RANKL monoclonal antibody) reduces skeletal events and is NOT cleared renally, making it the bone agent of choice in renal impairment. It causes hypocalcemia (supplement calcium/vitamin D), ONJ, and rebound vertebral fractures if stopped abruptly.

BOARD TRAP

Denosumab is preferred over bisphosphonates in renal failure but causes HYPOCALCEMIA — supplement calcium/vitamin D and don't stop abruptly (rebound fractures).

14. Emergency corridor (recognize 4 crises)

WHAT YOU SEE

A central red 'EMERGENCY CORRIDOR / EM2-ALERT CORRIDOR' linking the alert bays.

WHAT IT ENCODES

The four myeloma emergencies to recognize instantly: hypercalcemic crisis, spinal cord compression, hyperviscosity syndrome, and acute renal failure (cast nephropathy/tumor lysis). Each has a distinct first action — saline, dexamethasone+MRI, plasmapheresis, and hydration/light-chain reduction respectively.

BOARD TRAP

Match each crisis to its first move: hyperCa = saline; cord = dex + MRI; hyperviscosity = plasmapheresis; AKI = hydrate + lower light chains/avoid nephrotoxins — mixing these up is the classic test error.